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Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science and Engineering
End Semester Examination

B.E. COE/CSE: Semester-VII **Course Name: Compiler Construction (UCS 802)**
December 17, 2025, at 2 PM **Time: 3 Hours M. Marks: 40**
Faculty: Shalini Batra, Rupali Bhardwaj, Chinmaya Panigrahi, Shashank Singh, Karun Verma

a) ***Attempt any 4 out of 7 questions.***
b) ***Attempt parts of one question in sequence.***
c) ***Draw neat diagrams wherever required. Pencil should only be used to draw diagrams.***
d) ***Please indicate the sheet number for each attempted question on the cover page of your answer sheet.***

Q1 Consider the following translation scheme for signed binary numbers grammar G = {{N, S , L, B}, N, {+, −,0,1}, P}, where P is set of productions having elements:

$N \rightarrow S\ \{\boldsymbol{L.p = 0}\}\ L\ \{\boldsymbol{if\ (S.n)\ N.v = -L.v\ else\ N.v = L.v}\}$
$S \rightarrow\ +\ \{\boldsymbol{S.n = FALSE}\}$
$S \rightarrow\ -\{\boldsymbol{S.n = TRUE}\}$
$L \rightarrow \{\boldsymbol{B.p = L.p}\}\ B\ \{\boldsymbol{L.v = B.v}\}$
$L \rightarrow \{\boldsymbol{L_1.p = L.p + 1}\}\ L_1\ \{\boldsymbol{B.p = L.p}\}\ B\ \{\boldsymbol{L.v = L_1.v + B.v}\}$
$B \rightarrow\ 0\ \{\boldsymbol{B.v = 0}\}$
$B \rightarrow 1\ \{\boldsymbol{B.v = 2^{B.p}}\}$

Here v, n be synthesized attributes, and p an inherited attribute for the above grammar.

a) Using the above grammar G, write the rightmost derivation for the string $-\mathbf{101011}$. 4
b) With the translation scheme evaluate the binary expression into its equivalent decimal value, by showing the parse tree. (CO3, L5) 6

Q2 a) Differentiate between 4
i. Synthesized and Inherited attributes.
ii. Syntax tree and Directed Acyclic Graphs. (CO3, L3)

b) Find error in the below statement. What will phrase level recovery do in this scenario? 2

```
if (x > y {z = 5;}
```

(CO1, L2)

c) Differentiate between SLR, LR(1) and LALR parsers with a suitable example. Which of these parsers is used in practice for various compilers? (CO2, L3) 4

Q3 Consider the grammar G = {{E, T, F, M, A}, E, {+, −, ÷, ×, **id**, (,)}, P}, where P is set of productions having elements:

$E \rightarrow E\ M\ T \mid T$
$T \rightarrow T\ A\ F \mid F$
$F \rightarrow \mathbf{id} \mid (E)$
$M \rightarrow \div \mid \times$
$A \rightarrow + \mid -$

a) Remove Left recursion from the grammar. 2
b) Compute the FIRST and FOLLOW and derive the LL(1) parsing table. 6
c) Show the parsing operations for the string "**id * id + id**". (CO2, L4) 2

Q4 a) Consider the expression $((a + b) \times c) + (d \times (e + f))$
Assume that machine has only two registers (R_0, R_1).
i. Draw the syntax tree for the given expression. 2
ii. Use Sethi–Ullman algorithm to identify number of registers are required to evaluate the whole expression. 4

b) Draw the control flow graph for following code and identify how many basic blocks and edges does the control graph have? 4

```
i = 0;
while (i < n) {
if (a[i] > 0)
        sum = sum + a[i];
        else
                sum = sum - a[i];
        i = i + 1;
}
```

(CO4, L4)

Q5 a) What is error recovery in a compiler? Why is it needed? (CO1, L2) 2

b) Express the following expression in the form of syntax tree, quadruples, triples and indirect triples.
$(a + b) * (c + d) - (a + b + c)$ (CO3, L4) 4

c) Write three address code for the following C program fragment 4

```
While (p<1 and r> s) do
        If p = 1 then
                q = q+1
        Else
                While (p<=5) do
                        p = p+3
```

(CO3, L4)

Q6 a) Consider a regular expression $ab(a|b)bb^*$

i. Using the syntax tree method, draw the annotated syntax tree. Find the ***firstpos***, ***lastpos*** and ***followpos***. 5

ii. Construct the DFA for the above regular expression using the syntax tree method. 3

b) Left factor the following grammar

$S \rightarrow bSSbS \mid bSaSb \mid bba \mid b \mid a$ (CO2, L4) 2

Q7 a) Explain the concept of loop optimization in compiler design. Additionally, describe how control flow analysis is performed using basic blocks within a program flow graph. 4

b) Explain various storage allocation strategies with suitable examples. (CO3, L2) 6

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